Your Project: Final Version

What you're building: An ML system that predicts the weekly direction of Brent crude (buy, hold, sell) using an ensemble of Random Forest, XGBoost and LightGBM. Features are drawn from 8 data sources, evaluated with walk-forward backtesting so there is no data leakage. SHAP values show which features drive predictions and when. The output is a live Streamlit dashboard.

Target: will Brent close higher or lower 5 trading days from now? Buy above +1.5%, Sell below −1.5%, Hold in between. The threshold will be tuned and justified.

The features:

- Brent price and technical indicators (returns, RSI, MACD, Bollinger Bands, ATR, moving averages, futures basis)
- EIA US crude inventories and production
- OPEC production and Baker Hughes rig count
- Macro: DXY, VIX, S&P 500, 10-year yield, CPI
- Global manufacturing PMI
- CFTC speculative positioning
- AI-GPR indices (the baseline geopolitical features)
- Your own LLM-generated supply-risk scores, per producer country and chokepoint (the novel part)
- JWC war-risk insurance listings (also novel, not LLM-derived)

The research question

Do oil-supply-specific, country- and chokepoint-level features predict Brent's weekly direction better than a model already using the full AI-GPR suite?

Model A — standard features only

Model B — A + full AI-GPR suite

Model C — B + your LLM scores + JWC listings

Then you can answer two questions:

- Does geopolitical risk information help at all? (B vs A)
- Do your features add value on top of AI-GPR? (C vs B)

Compared using: walk-forward backtesting, SHAP, and a Diebold-Mariano significance test.

Case study: the 2026 Iran/Hormuz crisis, scored with an LLM whose training cutoff predates the crisis (exact cutoff date to be confirmed against OpenAI's documentation before this goes in the final paper).

The novelty

The AI-GPR (Iacoviello & Tong, 2026) is the most recent LLM-based geopolitical risk index. It publishes country risk by event type (e.g. Iran sanctions, Iran military conflict), but these capture any conflict or any sanctions, not whether oil supply is threatened. Its oil index is only split by region, so Saudi Arabia, Iran and Iraq sit in one Middle East bucket. It has no chokepoint series, and it was built to measure risk, not tested as a forecasting input for Brent.

Cao et al. (2026) extract per-country entity stances and event types from geopolitical news using a similar LLM approach, and predict Brent price level. However, their scores capture general political risk rather than oil-supply-specific risk, they use a fixed chronological train/validation/test split rather than walk-forward backtesting, and their framework includes no chokepoint features or insurance-market signals. This project extends that direction by asking the narrower supply question, adding chokepoints and JWC listings, and evaluating on weekly directional forecasting against a rigorous baseline.

This project:

1. Builds oil-supply-specific scores per producer country
2. Builds chokepoint-specific scores for Hormuz, Suez and Bab-el-Mandeb
3. Adds JWC insurance listings as a feature
4. Tests whether these additions improve weekly Brent direction forecasts beyond a model already using the full AI-GPR suite

The individual ideas are not entirely new. Analysts already watch chokepoint risk and war-risk insurance. The contribution is the combination and the test. To the author's knowledge, based on the literature reviewed so far, no study has done this.

Limitations to state up front

- **Hindsight bias:** an LLM scoring old news knows how events ended. Use a date-anchored prompt like the AI-GPR authors did, and admit it doesn't fully remove the bias.
- **Short case study:** the 2026 crisis gives only a few dozen weekly observations, so the Diebold-Mariano test will have low power there.
- **Frequency mismatch:** the AI-GPR headline index is daily, but its country-level breakdown is only published monthly, so those country series get forward-filled to weekly.
- **Source difference:** AI-GPR uses NYT, Washington Post and Chicago Tribune. Your scores come from GDELT, so the two models differ in news source as well as feature granularity. Acknowledge this rather than trying to match sources — matching would mean losing GDELT's global coverage.
- **JWC sparsity:** listings change rarely. Count the changes before committing to the feature, since outside crisis periods it may behave more like a crisis-period flag than a genuine weekly signal.

- **Data availability:** confirm the regional/country-level Oil GPR series are actually downloadable from the AI-GPR companion site before the plan is finalised. If they're not public, Model B falls back to the headline AI-GPR and Oil GPR indices only.

The argument to examiners

- Existing geopolitical risk indices measure general risk by country or by broad region. They don't separate threats to oil supply at producer or chokepoint level, and they weren't built for forecasting. This project uses an LLM to build oil-supply-specific scores per producer and per chokepoint, adds insurance-market listings, and tests whether these add predictive value beyond a model already using the full AI-GPR suite. The 2026 Hormuz crisis provides a case study after the LLM's training cutoff.

One-sentence version (presentation, oral)

- I'm building an ML system to predict weekly Brent direction. The novel part is using an LLM to score oil-supply risk for individual producers and chokepoints, then testing whether that adds predictive value beyond the existing AI-GPR geopolitical indices.

1. Price & Technical (Brent futures data)

- Daily returns — 1d, 5d, 20d
- RSI (14)
- MACD
- Bollinger Bands
- ATR
- 20 / 50 / 200-day moving averages
- Futures basis (backwardation/contango)

2. US Supply (EIA API — free)

- Crude inventories (weekly)
- Crude production (weekly)

3. Global Supply (OPEC reports — free)

- OPEC total production (monthly)
- Baker Hughes rig count (weekly)

4. Macro (FRED — free)

- USD Index (DXY)
- US 10-year Treasury yield
- CPI / inflation

- VIX
- S&P 500

5. Demand Proxy (ISM / Markit)

- Global Manufacturing PMI

6. Speculative Positioning (CFTC — free)

- Net non-commercial futures positions (weekly)

7. Geopolitical — Global (Caldara & Iacoviello — free)

- GPR Index (monthly)

8. Geopolitical — Country-level (your LLM pipeline)

- Per-country scores: conflict, stability, sanctions risk, infra attack, supply disruption
- Per-chokepoint scores: blockage risk, traffic disruption

The literature

1. Kilian (2009) — "Not All Oil Price Shocks Are Alike"
Decomposes oil price moves into supply shocks, demand shocks, and precautionary demand (fear). Shows each type has a different economic effect. Gives you the conceptual language for why oil prices move — essential framing for the introduction and feature design.

2. Alquist, Kilian & Vigfusson (2013) — Oil Price Forecasting (Handbook chapter)
The definitive survey of how well different models forecast oil prices. Futures markets don't beat a random walk; structural models can. Justifies why an ML ensemble is worth trying and sets the benchmark you need to beat.

3. Baumeister & Kilian (2016) — Real-Time Oil Price Forecasting
Shows you can beat the random walk with a combination forecast. Directly sets your baseline and is one of the papers your DM test should be comparing against conceptually.

4. Caldara & Iacoviello (2022) — Geopolitical Risk (GPR) Index
Builds a text-based geopolitical risk index from newspaper counts. The intellectual ancestor of your GPT-4o news scoring — you cite this to show the idea has strong prior work in economics.

5. Iacoviello & Tong — AI-GPR Index
Extends GPR using ML/NLP instead of keyword counts. The closest prior work to what you're doing with GDELT + GPT-4o mini. Probably your most important "related work" citation.

6. Cao (2026) — ML for Oil Price Prediction
A recent ML paper on the same problem. Shows the field is active and your work is timely, and gives you a direct methodological comparison.

7. Cerqueira, Torgo & Mozetic (2020) — Evaluating Time Series Forecasting
Compares evaluation strategies for ML on time series. Backs up your walk-forward validation design over standard cross-validation.

8. Lundberg & Lee (2017) — SHAP
The paper that introduces SHAP values. Justifies your explainability method — examiners will ask why SHAP, this is the answer.

9. Diebold & Mariano (1995) — Comparing Predictive Accuracy
The DM test. Lets you say statistically whether Model B/C beat Model A, not just that the numbers look better.